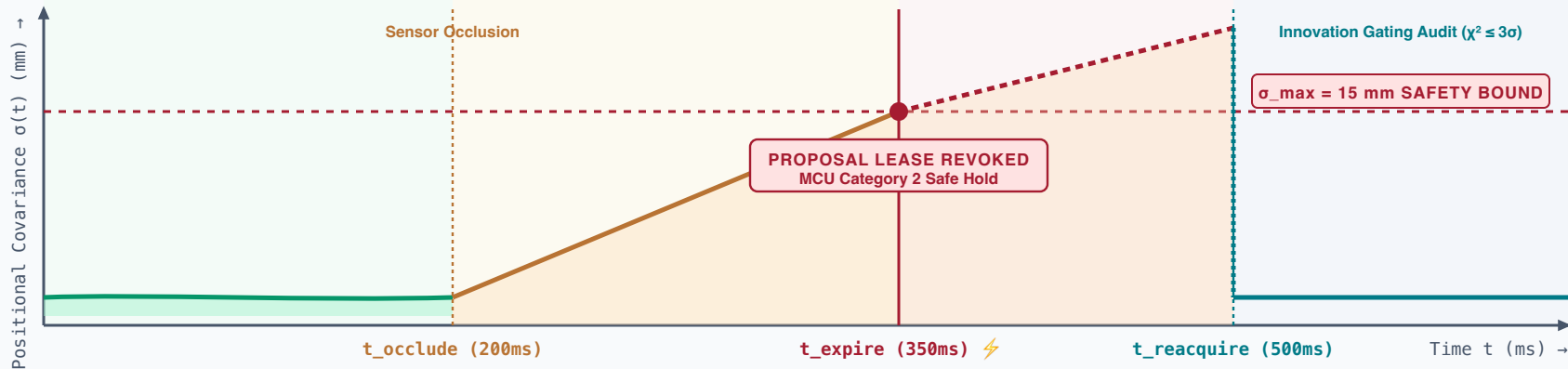


SPATIAL UNCERTAINTY EXPANSION, TTL LEASE EXPIRATION, & INNOVATION GATING

Autonomous revocation of unverified learned proposals when sensory occlusion expands spatial covariance beyond safety bounds



1. UNCERTAINTY DRIFT DYNAMICS

$$\sigma_{pos}(t) = \sigma_0 + v_{drift} \cdot \Delta t$$

- Without vision updates, dead reckoning accumulates drift and state noise.
- Moving target (0.1 m/s) adds **10 mm error every 100 ms**.
- State covariance matrix Σ expands.

2. HARDWARE LEASE REVOCATION

if ($\sigma_{pos} > \sigma_{max}$) RevokeAuthority()

- MCU audits spatial covariance every 1 ms tick in static SRAM.
- At $t_{expire} = 350\text{ ms}$, lease expires → revokes motor authority.
- Executes Category 2 safe hold.

3. χ^2 INNOVATION GATING AUDIT

$$d_M^2 = (z - \hat{z})^T S^{-1} (z - \hat{z}) \leq \gamma$$

- Upon return, Mahalanobis distance d_M audits candidate observation.
- **Valid ($\leq 3\sigma$):** Resets covariance, safely resumes trajectory execution.
- **Outlier ($> 3\sigma$):** Rejects spurious visual hallucination or reflection.